

❖ Console Input/Output

1. Write a program to take a student's **name, age, city, and percentage** as input and display them.
2. Take the **length** and **width** of a rectangle as input and calculate:
 - Area
 - Perimeter
3. Take the **price** and **quantity** of a product from the user and calculate:
 - Total Amount
 - GST (18%)
 - Final Bill
4. Take a student's marks **Python, Java, and SQL** as input and calculate:
 - Total
 - Average
 - Percentage

❖ Conditional Statements

1. Write a program to check whether a number is **positive, negative, or zero**.
2. Write a program to check whether a person is **eligible to vote**.
3. Write a program to find the **largest of two numbers**.
4. Write a program to find the **largest of three numbers** using if-elif-else.
5. Write a program to check whether a student has **passed or failed**. (Passing Marks = 35)

6. Create a simple **login system**.

- Correct username = "admin"
- Correct password = "1234"

Display:

- Login Successful
- Invalid Username
- Invalid Password

7. Write a program to determine whether a year is a **Leap Year**.

8. Create a simple calculator using **match-case**.

- Operations:
 - Addition
 - Subtraction
 - Multiplication
 - Division

9. Write a program using the **Ternary Operator** to determine whether a number is **Even or Odd**.

❖Loops

1. Print numbers from **1 to 100** using a for loop.
2. Print numbers from **100 to 1** using a while loop.
3. Print all **even numbers** between **1 and 100**.

4. Print all **odd numbers** between **1 and 100**.
5. Print the multiplication table of any number entered by the user.

- Ex

```
7 × 1 = 7
7 × 2 = 14
...
7 × 10 = 70
```

❖ Nested Loops

1. Print a **5 × 5** matrix of stars.

```
*****
*****
*****
*****
*****
```

2. Print the following pattern.

```
12345
12345
12345
12345
12345
```

3. Print the following pattern

```
11111
22222
33333
44444
55555
```

4. Print the following pattern

```
*****
****
***
**
```

5. Print the following pattern.

```
1
22
333
4444
55555
```

6. Print the following full pyramid.

```
  *
 ***
*****
*****
*****
```

7. Print the following Floyd's Triangle.

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```